

## 26.2 Variance inflation factors

Analysts usually want something coarser than the eigenvalues: one number per coefficient saying how much its precision has suffered. Chapter 6 supplied that number as a by-product of the Frisch–Waugh–Lovell theorem. This section states what it measures and what it does not, and extends it to a group of columns such as the indicators of a factor.

### 26.2.1 Definition and meaning

Throughout, the model contains an intercept,  $\mathbf{x}_j$  is one of the other columns, and  $\mathbf{X}_{(j)}$  is  $\mathbf{X}$  with  $\mathbf{x}_j$  removed, with projection  $\mathbf{M}_{(j)}$ . Write  $S_{jj} = \|\mathbf{x}_j - \bar{x}_j \mathbf{1}\|^2$  for the centred sum of squares of  $\mathbf{x}_j$ .

#### Definition 26.2.1. Variance inflation factor

Let  $R_j^2$  be the coefficient of determination of the least squares regression of  $\mathbf{x}_j$  on the columns of  $\mathbf{X}_{(j)}$ , which include  $\mathbf{1}$ , and suppose  $R_j^2 < 1$ . The **variance inflation factor** of  $\mathbf{x}_j$  is

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

and  $\text{TOL}_j = 1 - R_j^2 = 1/\text{VIF}_j$  is its **tolerance**.

The identity Eq. 6.6.2, proved with the Frisch–Waugh–Lovell theorem, reads  $\text{Var}(\hat{\beta}_j) = \sigma^2 \text{VIF}_j / S_{jj}$ . The next proposition collects what follows from it.

#### Proposition 26.2.1. What a variance inflation factor measures

- (a)  $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}/\text{VIF}_j$ , and  $\text{VIF}_j$  is the ratio of  $\text{Var}(\hat{\beta}_j)$  to  $\sigma^2/S_{jj}$ , the variance  $\hat{\beta}_j$  would have if the centred  $\mathbf{x}_j$  were orthogonal to every other centred column. By Exercise (A2),  $\text{VIF}_j \geq 1$ .
- (b)  $\text{VIF}_j$  depends on  $\mathbf{X}$  only through  $\mathbf{x}_j$  and  $\mathcal{C}(\mathbf{X}_{(j)})$ . It is unchanged when  $\mathbf{x}_j$  is replaced by  $a\mathbf{x}_j + b\mathbf{1}$  with  $a \neq 0$ , and when  $\mathbf{X}_{(j)}$  is replaced by any matrix with the same column space.
- (c) If  $\mathbf{R}$  is the correlation matrix of the non-intercept columns, then  $\text{VIF}_j = [\mathbf{R}^{-1}]_{jj}$ .
- (d) The  $t$  interval for  $\beta_j$  at any level is  $\sqrt{\text{VIF}_j}$  times as long, on average, as in the orthogonal design of (a), and the noncentrality of the  $F$  test of  $\beta_j = 0$  (the square of the  $t$  statistic's) is divided by  $\text{VIF}_j$ :

$$\frac{\beta_j^2}{\text{Var}(\hat{\beta}_j)} = \frac{\beta_j^2 S_{jj}}{\sigma^2} \cdot \frac{1}{\text{VIF}_j}.$$

*Proof.*

- (a) Since  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$ , the vector  $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$  is the residual of the regression of  $\mathbf{x}_j$  on  $\mathbf{X}_{(j)}$ , and its squared length is  $S_{jj}(1 - R_j^2)$  by the definition of  $R^2$  (Exercise (B1)). The variance ratio is Eq. 6.6.2 divided by  $\sigma^2/S_{jj}$ , which is the variance in the orthogonal case by Proposition 7.2.4 applied to the model with centred columns.
- (b) By (a),  $1/\text{VIF}_j = \|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2/S_{jj}$ , and  $\mathbf{M}_{(j)}$  depends only on  $\mathcal{C}(\mathbf{X}_{(j)})$ . Replacing

$\mathbf{x}_j$  by  $a\mathbf{x}_j + b\mathbf{1}$  multiplies the numerator by  $a^2$ , because  $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{1} = \mathbf{0}$ , and multiplies  $S_{jj}$  by  $a^2$ .

(c) This is Exercise (A1).

(d) The  $t$  interval has half-length  $t_{\alpha/2}(n-p) s \sqrt{\text{VIF}_j / S_{jj}}$ , and  $s$  has the same distribution in both designs. The noncentrality is  $\beta_j^2 / \text{Var}(\hat{\beta}_j)$  with the variance from (a) (Theorem 11.6.1).

■

By (b) the factor is scale-free: a property of the angle between the centred  $\mathbf{x}_j$  and the space of the other centred columns, whose squared cosine is  $R_j^2$  (Theorem 6.7.2). By (a), with the spread per observation fixed,  $\hat{\beta}_j$  is as precise as in an orthogonal design with  $n/\text{VIF}_j$  observations.

*Idea.* A variance inflation factor of 10 costs  $\hat{\beta}_j$  as much precision as discarding nine observations in ten from an orthogonal design with the same spread.

## 26.2.2 Reading variance inflation factors

Rules of thumb flag a regressor when  $\text{VIF}_j$  exceeds 10, or 5, or 4. The thresholds are arbitrary, and they measure the wrong thing. The factor compares the design with a hypothetical orthogonal design of the same size; it says nothing about whether the actual standard error is small enough for the purpose, which depends also on  $\sigma^2$  and on  $S_{jj}$ , hence on  $n$ . As  $n$  grows with the regressors drawn from a fixed population,  $\text{VIF}_j$  settles to a population value while  $S_{jj}$  grows like  $n$ , so a factor that is ruinous at  $n = 20$  can be irrelevant at  $n = 20,000$ . O'Brien (2007) makes this case in detail. A large factor is a reason to look at the interval, never a verdict by itself.

### Example. Large factors, precise estimate

The quarterly US macroeconomic series of Example (A macroeconomic design) run from 1959 to 2009,  $n = 203$ . Regress real personal consumption on real disposable income, population, the consumer price index, the money stock M1, the three-month Treasury bill rate and the unemployment rate, with an intercept. The variance inflation factors are

| Regressor      | income | population | CPI   | M1   | T-bill | unemployment |
|----------------|--------|------------|-------|------|--------|--------------|
| $\text{VIF}_j$ | 154.5  | 183.7      | 140.4 | 97.0 | 2.9    | 1.3          |

By every rule of thumb, four regressors are badly collinear. Yet the coefficient of income is 1.0659 with standard error 0.0210, a  $t$  statistic of 50.9: the standard error is  $\sqrt{154.5} \approx 12.4$  times what an orthogonal design would give, and still small, because income varies widely and consumption tracks it closely. (The errors are autocorrelated, so all the standard errors are too small, Chapter 21; the factors depend on  $\mathbf{X}$  alone.)

```
import numpy as np
import statsmodels.api as sm
```

```
def vifs(Z):
    """VIFs of the columns of Z (an intercept is always included): diagonal of R
```

```

 $\hat{\gamma}_{-1}$ ."""
    return np.diag(np.linalg.inv(np.corrcoef(Z, rowvar=False)))

macro = sm.datasets.macroddata.load_pandas().data
names = ["realdpi", "pop", "cpi", "m1", "tbilrate", "unemp"]
Z = macro[names].to_numpy()
fit = sm.OLS(macro["realcons"].to_numpy(), sm.add_constant(Z)).fit()
for name, v, b, se in zip(names, vifs(Z), fit.params[1:], fit.bse[1:]):
    print(f"{name:9s} VIF {v:7.1f}    estimate {b:9.4f}    se {se:7.4f}    t {b / se :6.1f}")

```

[Run this code online](#)

## 26.2.3 Collinearity that a reparameterization removes

Some large factors are produced by the parameterization, not by the data. Polynomial and interaction terms built from regressors whose origin lies far from the data are the standard case. Such collinearity is sometimes called *nonessential*.

### Example. A quadratic far from the origin

Take  $n = 21$  equally spaced values  $x_i = 10, 10.5, \dots, 20$  and the quadratic model  $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$ . Over this range  $x$  and  $x^2$  are almost perfectly correlated:  $R^2 = 0.9920$ , and both have  $VIF = 124.6$ . Rewrite the model in the centred regressors  $x - 15$  and  $(x - 15)^2$ . The fitted values are identical, and both variance inflation factors are now exactly 1, because the design is symmetric about 15 and an odd function of  $x - 15$  is orthogonal to an even one once both are centred.

What changed is the question. In the raw coding  $\beta_1$  is the slope of the mean function at  $x = 0$ , ten units below the data, with standard deviation  $0.804\sigma$ . In the centred coding the coefficient of  $x - 15$  is the slope at  $x = 15$ , with standard deviation  $0.0721\sigma$ , and the raw coding estimates the same slope,  $\beta_1 + 30\beta_2$ , with exactly the same standard deviation. The large factors described the difficulty of a question, the slope at the origin, that the analyst probably did not mean to ask ([Exercise \(B2\)](#) derives 124.6 by hand).

```

x = np.linspace(10, 20, 21)                                # a regressor far from its
    origin

raw = np.column_stack([x, x**2])
cen = np.column_stack([x - x.mean(), (x - x.mean()) ** 2])
print("VIFs, raw      :", np.round(vifs(raw), 1))
print("VIFs, centred :", np.round(vifs(cen), 3))

X_raw = np.column_stack([np.ones_like(x), raw])
X_cen = np.column_stack([np.ones_like(x), cen])
G_raw, G_cen = np.linalg.inv(X_raw.T @ X_raw), np.linalg.inv(X_cen.T @ X_cen)
a_mean = np.array([0, 1, 2 * x.mean()])                    # slope at x = 15 in the raw
    coefficients
print("sd of the slope at 0 (raw b1)      :", round(np.sqrt(G_raw[1, 1]), 3))
print("sd of the slope at 15, raw coding  :", round(np.sqrt(a_mean @ G_raw @ a_mean
    ), 4))
print("sd of the slope at 15, centred (b1) :", round(np.sqrt(G_cen[1, 1]), 4))

```

[Run this code online](#)

## 26.2.4 Groups of columns

A factor enters the model through several indicator columns, a polynomial through several powers. The individual coefficients of such a block depend on the coding (Definition 8.5.1), and so do their variance inflation factors. What is intrinsic is the joint precision of the block, which the next definition, due to Fox and Monette (1992), measures.

For the rest of the section, split the non-intercept columns of  $\mathbf{X}$  into  $\mathbf{Z}_1$  ( $n \times q$ ) and  $\mathbf{Z}_2$  ( $n \times (k - q)$ ), let  $\tilde{\mathbf{Z}}_1$  and  $\tilde{\mathbf{Z}}_2$  be the centred versions, and let  $\tilde{\mathbf{M}}_2$  project onto  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ . Assume  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_1, \mathbf{Z}_2]$  has full column rank.

### Definition 26.2.2. Generalized variance inflation factor

The **generalized variance inflation factor** of the block  $\mathbf{Z}_1$  is

$$\text{GVIF}_1 = \frac{\det(\tilde{\mathbf{Z}}_1^\top \tilde{\mathbf{Z}}_1)}{\det(\tilde{\mathbf{Z}}_1^\top (\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1)}.$$

The **canonical correlations** between the two blocks are defined through orthonormal bases  $\mathbf{Q}_1$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_1)$  and  $\mathbf{Q}_2$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ : they are the singular values  $\rho_1 \geq \rho_2 \geq \dots \geq \rho_m$ ,  $m = \min(q, k - q)$ , of the  $q \times (k - q)$  matrix  $\mathbf{Q}_1^\top \mathbf{Q}_2$ . The largest,  $\rho_1$ , is the largest correlation between a linear combination of the centred columns of  $\mathbf{Z}_1$  and a linear combination of the centred columns of  $\mathbf{Z}_2$  (Exercise (C1)).

### Proposition 26.2.2. Properties of the generalized factor

Let  $\beta_1$  be the coefficients of  $\mathbf{Z}_1$ .

- (a)  $\det \text{Cov}(\hat{\beta}_1) = \text{GVIF}_1 \cdot \det \text{Cov}_\perp$ , where  $\text{Cov}_\perp = \sigma^2(\tilde{\mathbf{Z}}_1^\top \tilde{\mathbf{Z}}_1)^{-1}$  is the covariance  $\hat{\beta}_1$  would have if  $\mathcal{C}(\tilde{\mathbf{Z}}_1)$  were orthogonal to  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ . So  $\text{GVIF}_1$  is the squared ratio of the volumes of the joint confidence ellipsoids for  $\beta_1$  in the two designs.
- (b)  $\text{GVIF}_1 = \prod_{i=1}^m 1/(1 - \rho_i^2)$ .
- (c)  $\text{GVIF}_1$  is unchanged when  $\mathbf{Z}_1$  is replaced by  $\mathbf{Z}_1 \mathbf{A} + \mathbf{1f}^\top$  and  $\mathbf{Z}_2$  by  $\mathbf{Z}_2 \mathbf{B} + \mathbf{1g}^\top$ , for nonsingular  $\mathbf{A}$  and  $\mathbf{B}$  and any vectors  $\mathbf{f}$  and  $\mathbf{g}$ . In particular it does not depend on the coding of a factor. It is symmetric:  $\text{GVIF}_1 = \text{GVIF}_2$ .
- (d) If  $q = 1$ ,  $\text{GVIF}_1$  is the variance inflation factor of Definition 26.2.1. In general, with  $\mathbf{R}$  the correlation matrix of  $[\mathbf{Z}_1, \mathbf{Z}_2]$  partitioned conformably,  $\text{GVIF}_1 = \det \mathbf{R}_{11} \det \mathbf{R}_{22} / \det \mathbf{R}$ .

*Proof.*

- (a) Since  $\mathcal{C}(\mathbf{1}, \mathbf{Z}_2) = \mathcal{C}(\mathbf{1}) \oplus \mathcal{C}(\tilde{\mathbf{Z}}_2)$  with orthogonal summands, the projection onto  $\mathcal{C}(\mathbf{1}, \mathbf{Z}_2)$  is  $n^{-1} \mathbf{11}^\top + \tilde{\mathbf{M}}_2$  (Theorem 6.3.7), and the residual of  $\mathbf{Z}_1$  after it is  $(\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1$ . By Theorem 6.6.2 and Eq. 6.6.1,  $\text{Cov}(\hat{\beta}_1) = \sigma^2(\tilde{\mathbf{Z}}_1^\top (\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1)^{-1}$ . If  $\mathcal{C}(\tilde{\mathbf{Z}}_1) \perp \mathcal{C}(\tilde{\mathbf{Z}}_2)$ , then  $\tilde{\mathbf{M}}_2 \tilde{\mathbf{Z}}_1 = \mathbf{0}$  and the covariance is  $\text{Cov}_\perp$ . Take the ratio of determinants. By Theorem 12.2.1(b), the volume of the confidence ellipsoid is proportional to  $(\det \Sigma)^{1/2}$ , where  $\Sigma$  is the covariance matrix, with the same constant in both designs.
- (b) Centring annihilates  $\mathbf{1f}^\top$  and commutes with multiplication on the right, so replacing

$\mathbf{Z}_1$  by  $\mathbf{Z}_1\mathbf{A} + \mathbf{1f}^\top$  replaces  $\tilde{\mathbf{Z}}_1$  by  $\tilde{\mathbf{Z}}_1\mathbf{A}$ , and both determinants in [Definition 26.2.2](#) are multiplied by  $\det(\mathbf{A})^2$ . For the same reason  $\mathbf{Z}_2 \mapsto \mathbf{Z}_2\mathbf{B} + \mathbf{1g}^\top$  replaces  $\tilde{\mathbf{Z}}_2$  by  $\tilde{\mathbf{Z}}_2\mathbf{B}$ , which leaves  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ , and so  $\tilde{\mathbf{M}}_2$ , unchanged. We may therefore take  $\tilde{\mathbf{Z}}_1 = \mathbf{Q}_1$ . Then  $\tilde{\mathbf{M}}_2 = \mathbf{Q}_2\mathbf{Q}_2^\top$ , and with  $\mathbf{C} = \mathbf{Q}_1^\top\mathbf{Q}_2$ ,

$$\text{GVIF}_1 = \frac{\det(\mathbf{Q}_1^\top\mathbf{Q}_1)}{\det(\mathbf{Q}_1^\top\mathbf{Q}_1 - \mathbf{C}\mathbf{C}^\top)} = \frac{1}{\det(\mathbf{I}_q - \mathbf{C}\mathbf{C}^\top)}.$$

The eigenvalues of  $\mathbf{C}\mathbf{C}^\top$  are  $\rho_1^2, \dots, \rho_m^2$  and  $q-m$  zeros, so the determinant is  $\prod_i(1-\rho_i^2)$ .

- (c) The invariance was shown in the proof of (b). Symmetry follows from (b), because  $\mathbf{Q}_1^\top\mathbf{Q}_2$  and  $\mathbf{Q}_2^\top\mathbf{Q}_1$  have the same singular values.
- (d) For  $q = 1$  the ratio is  $S_{jj}/\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = \text{VIF}_j$  by [Proposition 26.2.1\(a\)](#). For the determinant form, let  $\mathbf{D}$  be the diagonal matrix with  $\mathbf{R} = \mathbf{D}[\tilde{\mathbf{Z}}_1, \tilde{\mathbf{Z}}_2]^\top[\tilde{\mathbf{Z}}_1, \tilde{\mathbf{Z}}_2]\mathbf{D}$ , with diagonal blocks  $\mathbf{D}_1, \mathbf{D}_2$ . By Theorem 1.6.3,  $\det \mathbf{R} = \det \mathbf{R}_{22} \det(\mathbf{R}_{11} - \mathbf{R}_{12}\mathbf{R}_{22}^{-1}\mathbf{R}_{21})$ , and the Schur complement is  $\mathbf{D}_1\tilde{\mathbf{Z}}_1^\top(\mathbf{I} - \tilde{\mathbf{M}}_2)\tilde{\mathbf{Z}}_1\mathbf{D}_1$ . Since  $\mathbf{R}_{11} = \mathbf{D}_1\tilde{\mathbf{Z}}_1^\top\tilde{\mathbf{Z}}_1\mathbf{D}_1$ , the factors  $\det(\mathbf{D}_1)^2$  cancel in  $\det \mathbf{R}_{11} \det \mathbf{R}_{22} / \det \mathbf{R}$ , which leaves  $\text{GVIF}_1$ . ■

A generalized factor inflates a  $q$ -dimensional volume, so for comparison with single columns Fox and Monette suggest  $\text{GVIF}_1^{1/(2q)}$ , the geometric mean of the factors by which the axes of the confidence ellipsoid lengthen; for  $q = 1$  it is  $\sqrt{\text{VIF}_j}$ .

#### Example. A factor that explains a regressor

Grunfeld's panel (Example (Grunfeld's investment data)) has 220 firm-years of gross investment, market value and capital stock for 11 firms. Regress investment on value, capital and the firm, entered as 10 indicator columns with the first firm as reference. The generalized factors are

| Term    | columns $q$ | GVIF  | GVIF <sup>1/(2q)</sup> |
|---------|-------------|-------|------------------------|
| value   | 1           | 18.32 | 4.280                  |
| capital | 1           | 2.04  |                        |
| firm    | 10          | 23.61 | 1.171                  |

The canonical correlations between the firm block and the pair (value, capital) are 0.9694 and 0.5459, and  $1/((1 - 0.9694^2)(1 - 0.5459^2)) \approx 23.6$  as [Proposition 26.2.2\(b\)](#) requires. Market value is largely a characteristic of the firm. With the firm indicators in the model, the coefficient of value is identified by movements within firms (the within estimator of [Section 6.6](#)), and the part of value that neither firm nor capital explains is  $1/18.32$ , about 5.5 per cent, of its variation.

The ordinary factors of the ten indicator columns range from 1.8 to 16.8; with the last firm as reference the largest is 12.7. The generalized factors are the same under every coding, including sum-to-zero coding ([Proposition 26.2.2\(c\)](#)).

```
import pandas as pd
grun = sm.datasets.grunfeld.load_pandas().data
D = pd.get_dummies(grun["firm"], drop_first=True).to_numpy(dtype=float) # 10
    indicator columns
```

```

Zg = np.column_stack([grun["value"], grun["capital"], D])

def gvif(Z, block):
    """det(R_11) det(R_22) / det(R) for the columns in block against the rest."""
    R = np.corrcoef(Z, rowvar=False)
    rest = [k for k in range(Z.shape[1]) if k not in block]
    return (np.linalg.det(R[np.ix_(block, block)]) * np.linalg.det(R[np.ix_(rest,
rest)]))
        / np.linalg.det(R))

firm = list(range(2, Zg.shape[1]))
print("GVIF value :", round(gvif(Zg, [0]), 2))
print("GVIF capital:", round(gvif(Zg, [1]), 2))
print("GVIF firm :", round(gvif(Zg, firm), 2), " per dimension:", round(gvif(Zg,
firm) ** (1 / 20), 3))
print("VIFs of the firm indicators:", np.round(vifs(Zg)[2:], 1))

```

[Run this code online](#)

## 26.2.5 Exercises

### 26.2.5.1 A. Check your understanding

#### Exercise. A1

Which values of  $R_j^2$  correspond to variance inflation factors of 4, 5 and 10? By what factor does each multiply the length of a confidence interval for  $\beta_j$ , relative to the orthogonal design of [Proposition 26.2.1\(a\)](#)?

#### Exercise. A2

For a model with an intercept and two regressors with sample correlation  $r$ , show that  $VIF_1 = VIF_2 = 1/(1 - r^2)$ . What does [Proposition 26.2.1\(a\)](#) say about the design of Example (Two regressors that move together)?

### 26.2.5.2 B. Practice

#### Exercise. B1

Show that appending a column  $\mathbf{w} \notin \mathcal{C}(\mathbf{X})$  to  $\mathbf{X}$  cannot decrease the variance inflation factor of any column already in the model, and that it leaves  $VIF_j$  unchanged iff the residual of  $\mathbf{w}$  on  $\mathbf{X}_{(j)}$  is orthogonal to the residual of  $\mathbf{x}_j$  on  $\mathbf{X}_{(j)}$ . Is  $\text{Var}(\hat{\beta}_j)$  also nondecreasing?

*Solution.* Appending a column enlarges  $\mathcal{C}(\mathbf{X}_{(j)})$ , so  $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2$  cannot increase, and by [Proposition 26.2.1\(a\)](#)  $VIF_j$  cannot decrease. Let  $\tilde{\mathbf{w}} \neq \mathbf{0}$  be the residual of  $\mathbf{w}$  on  $\mathbf{X}_{(j)}$ . The enlarged space is  $\mathcal{C}(\mathbf{X}_{(j)}) \oplus \mathcal{C}(\tilde{\mathbf{w}})$  with orthogonal summands, so the squared residual of  $\mathbf{x}_j$  drops by  $(\tilde{\mathbf{w}}^\top \mathbf{x}_j)^2 / \|\tilde{\mathbf{w}}\|^2$ , and  $\tilde{\mathbf{w}}^\top \mathbf{x}_j = \tilde{\mathbf{w}}^\top (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ . So  $VIF_j$  is unchanged iff the two residuals are orthogonal. Since  $S_{jj}$  does not change,  $\text{Var}(\hat{\beta}_j) = \sigma^2 VIF_j / S_{jj}$  is nondecreasing too. This is the variance half of Theorem 19.1.3.

**Exercise. B2**

Let  $x_i = m + u_i$ , where the  $u_i$  are symmetric about zero. Write  $v$  for the mean of the  $u_i^2$  and  $w$  for the variance of the  $u_i^2$  (both with divisor  $n$ ). Show that the variance inflation factor of  $x$  and of  $x^2$  in the quadratic model is

$$\text{VIF} = 1 + \frac{4m^2v}{w}.$$

Check the value in [Example \(A quadratic far from the origin\)](#).

*Solution.* With one other regressor,  $\text{VIF} = 1/(1 - r^2)$ , where  $r$  is the correlation of  $x$  and  $x^2$  ([Exercise \(A2\)](#)). Since  $x^2 = m^2 + 2mu + u^2$ , the covariance of  $x$  and  $x^2$  is that of  $u$  and  $2mu + u^2$ , which is  $2mv$  because the third moment of  $u$  vanishes. The variance of  $x^2$  is  $4m^2v + w$ , since  $u$  and  $u^2$  are uncorrelated. So  $r^2 = 4m^2v^2/(v(4m^2v + w)) = 4m^2v/(4m^2v + w)$  and  $1/(1 - r^2) = 1 + 4m^2v/w$ . In the example,  $m = 15$  and  $u$  runs over  $-5, -4.5, \dots, 5$ :  $v = 192.5/21 \approx 9.167$ , the mean of  $u^4$  is  $3166.6/21 \approx 150.79$ , so  $w \approx 150.79 - 84.03 = 66.76$  and  $\text{VIF} \approx 1 + 900 \cdot 9.167/66.76 \approx 124.6$ .

**Exercise. B3**

Let  $\mathbf{x}_j$  be a column of the block  $\mathbf{Z}_1$ . Show that  $\text{GVIF}_1 \geq 1/(1 - R_{j|2}^2)$ , where  $R_{j|2}^2$  is the coefficient of determination of  $\mathbf{x}_j$  regressed on  $\mathbf{1}$  and  $\mathbf{Z}_2$  alone. Why is there no such inequality between  $\text{GVIF}_1$  and the ordinary  $\text{VIF}_j$  computed in the full model?

*Solution.*  $R_{j|2}$  is the correlation between the centred  $\mathbf{x}_j$  and its projection on  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ , which is a correlation between a vector  $\mathbf{Q}_1\mathbf{s}$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_1)$  and a vector  $\mathbf{Q}_2\mathbf{t}$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ , with  $\mathbf{s}$  and  $\mathbf{t}$  unit vectors. That correlation is  $\mathbf{s}^\top \mathbf{Q}_1^\top \mathbf{Q}_2 \mathbf{t} \leq \rho_1$  (Proposition 1.8.2; this is the first part of [Exercise \(C1\)](#)). Hence  $\text{GVIF}_1 \geq 1/(1 - \rho_1^2) \geq 1/(1 - R_{j|2}^2)$  by [Proposition 26.2.2\(b\)](#). The ordinary  $\text{VIF}_j$  regresses  $\mathbf{x}_j$  also on the other columns of its own block, which can explain it well even when  $\mathbf{Z}_2$  cannot. The indicators of a factor with a small reference level are an example: each is nearly determined by the others.

**26.2.5.3 C. Going deeper****Exercise. C1**

Show that  $\rho_1 = \max\{\cos \angle(\mathbf{u}, \mathbf{v}) : \mathbf{u} \in \mathcal{C}(\tilde{\mathbf{Z}}_1), \mathbf{v} \in \mathcal{C}(\tilde{\mathbf{Z}}_2), \mathbf{u}, \mathbf{v} \neq \mathbf{0}\}$ . More generally, show that the generalized eigenvalues of  $\text{Cov}(\hat{\boldsymbol{\beta}}_1)$  relative to  $\text{Cov}_\perp$ , the roots  $\theta$  of  $\det(\text{Cov}(\hat{\boldsymbol{\beta}}_1) - \theta \text{Cov}_\perp) = 0$ , are  $1/(1 - \rho_i^2)$ ,  $i \leq m$ , together with  $q - m$  ones. Deduce that for every  $\mathbf{a} \neq \mathbf{0}$  the ratio  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_1)/\mathbf{a}^\top \text{Cov}_\perp \mathbf{a}$  lies between 1 and  $1/(1 - \rho_1^2)$ .

*Solution.* Write  $\mathbf{u} = \mathbf{Q}_1\mathbf{s}$  and  $\mathbf{v} = \mathbf{Q}_2\mathbf{t}$  with unit  $\mathbf{s}, \mathbf{t}$ . Then  $\cos \angle(\mathbf{u}, \mathbf{v}) = \mathbf{s}^\top \mathbf{Q}_1^\top \mathbf{Q}_2 \mathbf{t}$ , whose maximum over unit vectors is the largest singular value of  $\mathbf{Q}_1^\top \mathbf{Q}_2$  (Proposition 1.8.2). For the second part, take  $\tilde{\mathbf{Z}}_1 = \mathbf{Q}_1$  by [Proposition 26.2.2\(c\)](#), which changes both matrices by the same congruence and so leaves the generalized eigenvalues unchanged. Then  $\text{Cov}_\perp = \sigma^2 \mathbf{I}$

and  $\text{Cov}(\hat{\boldsymbol{\beta}}_1) = \sigma^2(\mathbf{I} - \mathbf{C}\mathbf{C}^\top)^{-1}$ , whose eigenvalues are the stated ones. The ratio in the last part is a ratio of quadratic forms, bounded by Corollary 1.7.8, so it lies between the smallest and the largest generalized eigenvalue.